

Le-Chris Wang

Curriculum Vitae



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EDUCATION

- AUG. 2025 – MAY 2030 **Princeton University, Princeton, NJ, USA**
PhD in Astrophysical Sciences
Advisor: Prof. Joshua Winn, Dr. Shreyas Vissapragada
- AUG. 2021 – MAY 2025 **Johns Hopkins University, Baltimore, MD, USA**
BSc in Computer Science and Physics, Both with Departmental Honors
Minor: Applied Mathematics & Statistics and Pure Mathematics
Advisor: Prof. Kevin Schlaufman, Prof. David Sing

PUBLICATIONS

First-authored/equivalent Publications

7. **Wang, L. C.**, Sagnick Mukherjee, Stephen P. Schmidt et al., "Asymmetric Aerosol Distribution on the Terminators of the Warm Saturn WASP-69 b Revealed by JWST NIRISS/SOSS" Submitted to *ApJL*.
6. **Wang, L. C.** & Schlaufman, K. C., "Elemental Abundance Trends with Condensation Temperature are Unrelated to Planet Formation" Submitted to *ApJ*.
5. **Wang, L.-C.** & Winn, J. N., "How Many Transiting Giant Planets Can JWST Search for Moons and Rotational Oblateness?" *ApJL* (2026).
4. **Wang, L.-C.**, Sing, D. K., & Rustamkulov, Z., "A Comprehensive Analysis of the Panchromatic Transmission Spectrum of the Hot-Saturn WASP-96 b: Nondetection of Haze, Possible Sodium Limb Asymmetry, Stellar Characterization, and Formation History" *AJ* (2026).
3. Liu, R.*, **Wang, L.-C.***, Rustamkulov, Z., & Sing, D. K., "Unveiling the atmosphere of the super-Jupiter HAT-P-14b with JWST NIRISS and NIRSpec" *AJ* (2025, *: Co-first author).
2. Gou, X.*, Pan, X.*, **Wang, L.-C.***, "General Relativity Testing in Exoplanetary Systems" *IOP Conf. Ser.: Earth Environ. Sci.* (2021). (*: Equal contributions).
1. Zheng, Y., Wang, X., **Wang, L.-C.*** et al., "Test of Bell's and Mermin's inequalities on Quantum Computer" *2020 2nd International Conference on Information Technology and Computer Application* (2020) (*: Corresponding Author).

Coauthoed Publications

8. Katherine A. Bennett et al., incl. **Wang, L.-C.**, "A JWST NIRISS/SOSS Transmission Spectrum of the Exo-Neptune HAT-P-11 b Reveals Aerosols and a High Metallicity In Agreement with the Mass-Metallicity Trend", submitted to *AAS Journals*.
7. Ashtari, R. et al., incl. **Wang, L.-C.**, "A Clearer View of HAT-P-1 b: JWST NIRSpec G395H Reveals a Cloud-free Atmosphere of Water, Carbon Dioxide, and Possibly Hydrogen Sulfide", submitted to *AAS Journals*.
6. Sing, D. K. et al., incl. **Wang, L.-C.**, "A JWST transiting survey of FGK stellar limb darkening: empirical evidence for quadratic laws and atmospheric model comparisons", submitted to *AAS Journals*.

5. Schmidt, S. P. et al., incl. **Wang, L.-C.**, "Mitigating Charge Migration in JWST NIRISS Reveals That KELT-7 b is a Metal-enriched Ultra-hot Jupiter Orbiting a Young Metal-rich Star" preprint, submitted to *AAS Journals*.
4. Mukherjee, S. et al., incl. **Wang, L.-C.**, "Cloudy mornings and clear evenings on a giant extrasolar world" *Science* (2026).
3. Schmidt, S., MacDonald, R., Tsai, S., Radica, M., **Wang, L.-C.** et al., "A Comprehensive Reanalysis of K2-18 b's JWST NIRISS+NIRSpec Transmission Spectrum" *AJ* (2025).
2. Wang, G., incl. **Wang, L.-C.**, "A Revised Density Estimate for the Largest Planet, HAT-P-67 b" *AJ* (2025).
1. Chen, H., Clement, M. S., **Wang, L.-C.**, & Gu, J. T., "Born Dry or Born Wet? A Palette of Water Growth Histories in TRAPPIST-1 Analogs and Compact Planetary Systems" *ApJL* (2025).

AWARDS & FELLOWSHIPS

2025	First-Year Fellowship in the Natural Sciences and Engineering <i>One-year awards for doctoral students newly admitted by a degree program in Natural Sciences or Engineering for the first-year of study.</i>
2024	IDIES Summer Student Fellowship <i>\$6,000, JHU, competitive research fellowship awarded to 5 undergraduates conducting data-intensive research.</i>
2024	ΣΠΣ Physics Honor Society <i>Awarded for outstanding academic achievement in physics and astronomy.</i>
2023	Summer Provost's Undergraduate Research Award <i>\$6,000, JHU, competitive research fellowship (36 out of 400+ applicants).</i>
2023	Dean's ASPIRE Grants <i>\$2,474, JHU, competitive research fellowship; awarded to 10 undergraduates per year.</i>
2022	Hophacks 2nd place <i>Hopkins's premier 36-hour hackathon. 2/40. \$512 prize.</i>
2022	Quest2Learn Most Innovative Platform to Help with Learning <i>Awarded for creating an application that helps with learning.</i>
2022	Bloomberg Distinguished Professor Fellowship <i>\$6,000, JHU, competitive research fellowship; awarded to 2 physics undergraduates.</i>
2021–2025	Dean's List <i>Excellence in academics. Awarded every semester (8/8).</i>

TALKS & PRESENTATIONS

DEC. 2025	Unveiling the Formation History of the Cloud-free Exoplanet <i>International Conference on Exoplanets and Planet Formation</i>
OCT. 2024	Planet Formation with N-body Simulations <i>2024 IDIES Annual Symposium</i>
APRIL 2024	FIREFLy-SOSS: Exoplanet Transit Light Curves Extraction Pipeline for JWST NIRISS-SOSS Observations <i>Departmental Undergraduate Research Showcase, Johns Hopkins University, MD</i>
APRIL 2024	Is The Formation Of Planets The Cause of Solar Atypical Abundance Pattern? <i>Johns Hopkins University DREAMS Symposium</i>
APRIL 2024	Characterization of Cloud-free Hot-Saturn WASP-96b with Joint JWST, Hubble, VLT, and Spitzer Transmission Spectroscopy <i>Johns Hopkins University DREAMS Symposium</i>
JAN. 2024	Elemental Abundance Trends with Condensation Temperature are Unrelated to Planet Formation

JUNE 2023	243rd Meeting of the American Astronomical Society, New Orleans, LA Elemental Abundance Trends with Condensation Temperature are Unrelated to Planet Formation
JUNE 2023	Origins of Solar Systems Gordon Research Conference, Mount Holyoke College, MA Stellar Elemental Abundance Patterns: Implications for Planet Formation
AUG. 2022	No-PhD Journal Club, Johns Hopkins University, MD Optimizing JWST BOTS Transit Light Curve Fitting The Center for Astrophysics Research Experience, Johns Hopkins University, MD

TELESCOPE ALLOCATIONS

2026 Q3	Apache Point Observatory, ARCTIC, 2 nights <i>Star Monitoring and Refining Ephemerides for Exoplanets Favorable for Oblateness and Exomoons</i> PIs: Wang, L.-C. & Wang, G.
2024 Q4	Apache Point Observatory, ARCSAT, 14 nights <i>Complementary Flare Monitoring of the Enigmatic Triple Red Dwarf LTT 1445ABC</i> PIs: Rustamkulov, Z., Bennett, K., CoIs incl. Wang, L. C.
2024 Q3	Apache Point Observatory, ARCTIC, 3 nights <i>Synergistic Cool Star Monitoring: Characterization of Starspots</i> PIs: Rustamkulov, Z., Allen, N., Wang, L. C. , Wang, G.

TEACHING APPOINTMENTS

2024 SPRING	Teaching Assistant, AS.171.108 General Physics II (Undergraduate, 23 students)
2023 FALL	Teaching Assistant, AS.171.107 General Physics I (Undergraduate, 46 students)
2023 SPRING	Teaching Assistant, AS.171.101 General Physics I (Undergraduate, 46 students)
2022 FALL	Teaching Assistant, AS.171.101 General Physics I (Undergraduate, 23 students)

SKILLS

PROGRAMMING	Python, C/C++, Java, Assembly, Fortran, Matlab, R, HTML, CSS, JavaScript, Bash, rust, go, React, flask, Node.js
DATA SCIENCE	sql, MySQL, PostgreSQL, Pytorch, TensorFlow, OpenCV, docker
ASTRONOMY SOFTWARES	DS9, Siril, MESA (stellar structure), Rebound (N-body), Mercury (N-body), petitRADTRANS (atmospheric retrieval)
OBSERVATION EXPERIENCE	The Morris W. Offit Telescope (half-meter telescope at JHU), ARC 3.5m telescope at Apache Point Observatory
LANGUAGES	English, Chinese, French
OTHER	L ^A T _E X, Git, Slurm, Mathematica, JupyterLab, Adobe Lightroom, Adobe Photoshop, Blender, A Cappella, Marathon